

**THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS**

MSBA7028 Deep Learning

GENERAL INFORMATION	
Instructor: Prof. Xiao Lei Email: xlei@hku.hk Office: KK Leung 817 Phone: 3917 1033 Consultation times: By appointment Tutor: TBA Pre-requisites: MSBA7002 Business Statistics and MSBA7027 Machine Learning. (Basic linear algebra, basic statistics and probability, and basic programming experiences with Python.) Course Website: Moodle The following programming languages are used in this course: Python.	
COURSE DESCRIPTION	
Deep learning being the core of artificial intelligence, has become more and more important. This course provides a broad introduction to deep learning, including fundamental concepts, state-of-the-art algorithms for data analysis, and their application to real cases.	
COURSE OBJECTIVES	
1. To provide students a broad introduction to deep learning. 2. Understand state-of-the-art algorithms used in deep learning and learn how to apply these algorithms to solve problems with Python. 3. Gain proficiency in applying standard software packages and systems. 4. To provide students extensive hands-on programming and analysis experiences on practical problems.	
PROGRAMME LEARNING OUTCOMES	
PLO1: Acquisition and internalization of knowledge of the programme discipline PLO2: Application and integration of knowledge PLO3: Inculcating professionalism and leadership PLO4: Developing global outlook PLO5: Mastering communication skills	
COURSE LEARNING OUTCOMES	
Course Learning Outcomes	Aligned Programme Learning Outcomes

CLO1: Clearly identify and formulate the relevant problem		PLO1 PLO2	
CLO2: Select and use effective methods to address the problem		PLO2, 4	
CLO3: Use software tools to provide solution to the issue at hand CLO4: Communicate the solution effectively		PLO3, 5	
COURSE TEACHING AND LEARNING ACTIVITIES			
Course Teaching and Learning Activities		Expected contact hour	Study Load (% of study)
T&L1. Interactive lectures		30	25%
T&L2. Assignments, tutorials		60	50%
T&L 3 Self-study, group discussion		30	25%
Total		120	100%
Assessment Methods			
Brief Description (Optional)		Weight	Aligned Course Learning Outcomes
A1. Participation		20%	CLO1, 2, 3, 4
A2. Assignments		30%	CLO1, 2, 3, 4
A3. Final project		50%	CLO1, 2, 3, 4
Total		100%	
STANDARDS FOR ASSESSMENT			
Course Grade Descriptors			
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • High participation in discussions • Present arguments that have an element of originality • Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem solving skills • Excellent performance in assignments		
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Good participation in discussions • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the exams with accurate computation and good analytical and problem solving skills • Good performance in assignments		
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Some participation in discussions • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem solving skills • Acceptable performance in assignments		

D+, D	• Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Minimal or no participation in discussions • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem solving skills • Marginally acceptable performance in assignments
F	• Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Minimal or no participation in discussions • Present arguments poorly • Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem solving skills • Poorly performance in assignments
Assessment Rubrics for Participation	
A+, A, A-	• High participation in discussions • Always attend in-class discussions • Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • Present arguments that have an element of originality • Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	• Good participation in discussions • Often attend in-class discussions • Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Present arguments that go beyond the lecture and textbook • Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	• Some participation in discussions • Sometimes attend in-class discussions • Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Present arguments in a well-structure manner • Respect others and follow the class rules (no chatting and do not use cell phone)
D+, D	• Minimal or no participation in discussions • Rarely attend in-class discussions • Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Present arguments in a marginally acceptable manner • Respect others and follow the class rules (no chatting and do not use cell phone)
F	• Minimal or no participation in discussions • Almost never attend in-class discussions • Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Present arguments poorly • Behave poorly in class (often chatting with others, using cell phones, or being late)
Assessment Rubrics for Assignments and the Exam	
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Present arguments that have an element of originality • Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills

D+, D	• Demonstrate a minimum understanding of the concepts involved • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills
F	• Demonstrate a poor understanding of the concepts involved • Present arguments poorly • Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills
COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE	
Topic 1: Introduction to deep learning Topic 2: Neural network, multi-layer perceptron Topic 3: Convolutional neural network Topic 4: Computer vision Topic 5: Recurrent neural network Topic 6: Natural language processing Topic 7: Reinforcement learning Project presentation	
REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)	
Required textbooks: Dive into Deep Learning https://d2l.ai/ There is no one textbook that covers all the topics in this course. The instructor will provide materials on the course home webpage. The following book is recommended but not required. Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. http://www.deeplearningbook.org/	
MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE	
X conducting mid-term survey in additional to SETL around the end of the semester ☐ Online response via Moodle site ☐ Others: _____ (please specify)	
COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)	
An orderly learning environment is extremely important for this course. Disruptive behaviors are absolutely unacceptable. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Any such dishonesty will result in an F grade.	
ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)	
No late assignment will be accepted. Lecture notes and self-learning materials will be uploaded on Moodle. The instructor reserves all the rights to make necessary changes to the syllabus. These changes will be announced as soon as possible.	